DINKDATA

Design document

Product requirements, technical design and system design for DinkData, which turns PickleCam videos into per-player pickleball stats and highlight reels.

Version 1.10

21 October 2026

Owners: Robby and Marti

Prepared with Claude Code

Source: `docs/DESIGN.md`

About this document

The product requirements, technical design and system design for DinkData. This file is the single source of truth.

Version	1.10
Date	21 October 2026
Owners	Robby (`@roberto-cantero`) and Marti (`@probablynotmarti`)
Prepared with	Claude Code, after a review session with Robby and review comments from Marti
Replaces	`docs/DinkDataRebuildPlan.pdf` and every doc based on the `legacy/` prototype
Status	Final draft, reviewed by both owners in its PR. Nothing is built yet

Section 1 is the summary. Sections 2 and 3 explain why and what. Sections 4 to 7 explain how the system works. Sections 8 to 13 cover data, labelling, evaluation, compute, the repo and licensing. Section 14 is the roadmap and the setup phase, and section 15 lists risks. The glossary in section 4 defines every technical term.

Contents

1. Summary
2. Findings
3. Product requirements
4. Glossary
5. System architecture
6. The pipeline, stage by stage
7. Data contracts
8. Data, storage and privacy
9. Labelling and the gold set
10. Evaluation and targets
11. Compute and costs
12. Repository and development workflow
13. Licensing
14. Roadmap, setup phase and first issues
15. Risks and open questions
16. Sources

Appendices: A. Decision log · B. Questions for the PickleCam backend developer

1. Summary

DinkData takes a PickleCam recording of a doubles pickleball game and produces two things for every player in it: **stats** and a **highlight reel**. PickleCam cameras sit above courts at about 140 facilities and record thousands of games a week. For now DinkData runs on its own: a video goes in, stats and reels come out. Sending them with PickleCam's video emails comes later, once the accuracy targets are met.

DinkData is a standalone app first. The first goal is a prototype whose stats and reels are accurate and reliable enough for PickleCam to pay for, because players ignore or complain about wrong stats. We show it to PickleCam's CEO at the end of M3. The prototype is built on about 150 games downloaded by hand from the facility dashboard. `dink run` takes one of them and writes the report and each player's reel MP4 to a folder. Integration with PickleCam, automatic intake, delivery to players and production cost come in M4, after PickleCam approves the prototype, built around whatever PickleCam already runs (ADR 0018).

The v1 has two priorities, and each owner leads one.

- **Stats, led by Robby.** Serve in % and depth, return in % and depth, net and out errors per player, and winners per player. A landing map of serves, returns and out-errors comes almost free. DinkData doesn't report the score (ADR 0019).

- **Highlights, led by Marti.** A short video for each player, stitched automatically from their best rallies in the game.

The pipeline has three layers. Perception finds the court, the ball and the four players in every frame. Structure turns those detections into a game log of rallies, hits, bounces, who hit each ball and who won each rally. Outputs read the game log and produce stats, a report and the highlight reels.

Five ideas carry the design.

1. **The rules of pickleball are the strongest model.** A rally grammar decoder picks the most likely sequence of events inside a rally that obeys the rules. It corrects many missed and extra detections without training labels.
2. **Nothing depends on the players keeping score.** Many PickleCam players are beginners or play for fun, and they lose track of side-out scoring. Players know their own score and would spot any score that differs from theirs, so DinkData never reports it (ADR 0019). Comparing one serve with the next still hints at who won the rally in between, but only as weak evidence next to the ball. Telling the two partners apart matters for every per-player serve stat, so M0 measures it first.
3. **The cameras are fixed, so each one is calibrated once.** The calibration fits a real lens model, including wide-angle distortion, from court points, lines and the net. Each video also gets background images of the empty court, which gave ball trackers their biggest accuracy gain in an independent benchmark.
4. **Rules first, models second.** Every stage starts as a simple rule-based version, so the pipeline runs end to end early. People correct its suggestions in our labelling tool, and the corrections train the real models.
5. **Measure everything.** One command scores every stage against a hand-labelled answer key called the gold set. Every pull request shows its numbers, so AI agents can build safely.

The accuracy target is each player's count for each stat off by at most one in at least 90% of player-games. We measure accuracy over the stats we ship and always report coverage, the share of stats shipped. A stat with low confidence is left out, not shown wrong. During the beta, a person reviews every game.

After the prototype, cost per video matters, because PickleCam receives about 4,500 videos a day. Built the straightforward way, production would cost about $13,000 a month. The plan targets under $0.01 a video, about $900 to $2,000 a month at today's volume, through cheaper GPUs, processing only the play segments, a cheaper bandwidth tier, and storing results instead of videos (section 11).

Milestone	Ends with
Setup, done by this session	Docs, an empty package layout, CI and GitHub issues ready for other sessions
M0 Foundations	Data contracts, 150 hand-downloaded games in storage, stats tested on hand-written logs, a highlight renderer working from a hand-written rally list, partner identity measured, and labelling under way
M1 Ground truth	Calibrated cameras, the labelling tool, the gold set, the eval command, and a first end-to-end version producing a report and a reel
M2 Serve states and first reels	Serve states and rally winners meet their targets, rally-based reels per player, ball tracker v1
M3 Ball stats and full reels	Stats 2 to 5, the landing map and per-player reels built from winners and aces
M4 Integration and beta, after PickleCam approves the prototype	Automatic intake through PickleCam's endpoint, batch processing, reels played over PickleCam's video, automatic calibration, delivery to beta players with human review, and a measured production cost per video

M0 to M3 build the prototype. M3 ends with every v1 stat and reel meeting its gate on held-out facilities, and that is the version we show PickleCam's CEO. After M2 we decide whether an internal preview of the first reels is worth showing early.

2. Findings

2.1 The footage

These findings come from the PickleCam dashboard (five facilities and seven cameras, including one diagonal camera) and from the video host's API documentation, on 17 October 2026.

Finding	Consequence
About 140 facilities with 3 to 10 cameras each. One facility recorded 50 sessions in about 12 hours	Footage is plentiful. Labelling time is the bottleneck
A recording is usually one game of 11 to 14 minutes. Recordings stop at 45 minutes	Recordings can still contain warm-up, several games or a game cut off at 45 minutes
Videos are deleted 14 days after upload	We copy the games we need promptly
1080p H.264 with audio at about 4.4 Mbps, roughly 0.4 GB per 12-minute game. The nominal rate is 30 fps, with occasional dropped frames, from several camera models	All timing uses frame timestamps, never `frame / 30`
The host is Bunny Stream, with one library per facility and titles like `court<N>_<sessionId>`	We can list and download videos through its API
Wide-angle lenses bend the court lines. Corners are often out of frame, and diagonal cameras miss part of the near court	A plain 4-corner homography is wrong. Calibration must work from partial courts, and some bounces are not visible
On low cameras the near players' feet are often below the frame	Player positions need a fallback
Neighbouring courts appear in frame. A PickleCam watermark with a ball icon and a facility logo sit in the corners	Detections are filtered to our court, and the overlays are masked
Every recording lists the players' email addresses	Our data keeps opaque ids only
The audio carries music, talk, shoe squeaks and 3 to 8 neighbouring courts	We don't use audio, and we drop it when copying

Camera setup	Approximate share	What it looks like
Down-the-line, low (2 m or below)	64%	Behind one baseline with steep perspective. Near corners and feet are often cut off
Down-the-line, high (above 2 m)	16%	All corners usually visible, with less occlusion
Diagonal	20%	Off a corner. The net runs diagonally and much of the near court is out of frame

The shares are the team's estimates. The seven sampled cameras are consistent with them.

2.2 The current repo

The repo already has good tooling: uv, ruff, pyright, pre-commit with secret scanning, CI, CODEOWNERS and templates. It was built around a pre-AI prototype, though.

- `legacy/` is a fork of TrackNetV3, a badminton tracker. Its checkpoints and labels are lost. Its code assumes a CUDA GPU, loads whole videos into memory, assumes 1280×720 at 30 fps and measures everything in pixels. We tag it and delete it, and start from scratch.
- The court, stats and failure-log docs and the old plan PDF were written from the prototype. This document and short topic docs replace them.
- The event schema can't express a bounce that isn't visible, a game, a timestamp or a serve. Contract rules run only inside tests. `torch` is a core dependency, so even stats tests install gigabytes. `.gitignore` hides any folder named `test/` and any `*.sh` file. CI runs only on Linux. `LICENSE` is MIT and names only the TrackNetV3 author.
- Marti's open PR #26 sets up Matt Pocock's agent skills. Section 12.2 builds on it.

2.3 Research

- We found no published computer-vision research on pickleball. Our thousands of games from fixed cameras are an advantage.
- For ball tracking, WASB, published at BMVC 2023, beat TrackNetV2 in its authors' tests, but its official code can't train. The independent RacketVision benchmark from 2025 covered about 435,000 frames of tennis, badminton and table tennis. TrackNetV3 with a background image and 4 input frames was best overall, and the background image alone cut pixel error by 54 to 61%.

- Event-spotting models find tennis hits and bounces to within a frame or two. A bounce classifier working on 2D ball paths reached an F1 of 0.92 on tennis, and the way it proposed candidates mattered more than the classifier.
- A tennis system that decoded ball events with a grammar of the rules had fewer false and missed events than detectors alone.
- PnLCalib calibrates sports cameras from points and lines and models lens distortion. One flat, partial court constrains a lens poorly, so lens parameters should be shared across cameras of the same model, and the net adds a known height.
- Video AI models such as Gemini give timestamps to about a second, too coarse for hits. SAM 2 and SAM 3 struggle with small fast balls. Audio hit detection fails with neighbouring-court noise.
- CVAT and Label Studio are the standard free labelling tools. Racket-sports teams often build a small custom tool for event labelling.

3. Product requirements

3.1 Users and outputs

Players get a video of their game but no numbers and no highlights. Competitors already analyse games from facility cameras. PB Vision takes video from PodPlay clubs, from Sporfie's recording pipeline, and since April 2026 from Save My Play cameras through DUPR, where players need no app or account. Our advantage is PickleCam's existing reach: cameras at about 140 venues recording thousands of games a week, with players' emails already attached. DinkData adds stats and reels to games PickleCam records anyway, so a player gets them without doing anything.

The v1 users are recreational doubles players at PickleCam venues, through a beta, and the two owners, who need tools to label, review and measure. Facilities, coaches and the PickleCam integration come later.

The input is one PickleCam recording, identified by an opaque `video_id`, with its camera id. For each game in the recording, DinkData writes these files.

- `events.json`, the game log. It stays internal, because its rally winners and servers would let anyone rebuild the score.
- `stats.json`, every stat for every player, with a confidence and a shipped flag. It stays internal too, and the report shows its shipped stats.
- `report.html`, a one-page report headed by the game number, date and length, with a card per player with a photo and their numbers, and the landing map. It shows no score.
- `highlights.json`, a reel for each player as a list of clips, and one reel MP4 per player rendered from it. In M4 a web player can play the clips from PickleCam's video instead, and MP4s are rendered only on request (ADR 0012).

Every player gets the whole-game report and every reel, and finds themselves by photo. The v1 covers doubles under any scoring system, because no stat depends on the score.

3.2 Stats (Robby)

Every stat is a pure function of `events.json`, with unit tests built from hand-written logs. `docs/STATS.md` will hold the full definitions and becomes the authority for the code.

Stat	Definition
Serve in %	Serves whose first bounce lands in the diagonal service court, divided by serves. A serve touching the kitchen line is a fault. Close calls go to review
Serve depth	Feet from the serve's first bounce to the receiver's baseline. Deep is 5 ft or less, medium 5 to 10 ft, short over 10 ft, to be confirmed on real data
Return in %	Returns whose first bounce lands in the serving team's half, divided by returns
Return depth	Feet from the return's bounce to the serving team's baseline, with the same buckets
Net and out errors	The player hit the shot that ended the rally, and it didn't cross the net or first bounced outside the court. Serve faults are counted separately

Stat	Definition
Winners	The last shot of a rally the player's team won, which no opponent touched
Landing map	A top-down court showing the player's serve and return bounces and the bounce spots of their out-errors

Serve faults come free from the same events. DinkData shows no number that players could check against the score they remember: no final score, games won, points scored or rallies won (ADR 0019). Later stats, in order unless the beta says otherwise, are dink success rate (a dink that lands in the opponents' kitchen and isn't an error), an all-shots heat map, unforced errors and forehand versus backhand. The owners approve any other stat first.

3.3 Highlights (Marti)

A highlight reel is a short MP4 for one player, built from clips of rallies in that game. PickleCam's own highlights are fixed 30-second clips that a command at the court creates, and few recordings have one. Nothing picks highlights automatically (`docs/DATA.md`). DinkData's reels need no action from the players.

DinkData scores every rally for each player and keeps the best ones. The v1 rules are below.

Moment	Score	Available from
A long rally the player took part in, by shot count or duration	Higher for longer rallies	M2, from rally start and end
A rally the player's team won on that player's serve	Medium	M2, from the rally winner
The final rally of the game	Bonus	M2
A winner hit by the player	High	M3, from hitter attribution
An ace: the player's serve that the receiver didn't return	High	M3
A long dink exchange	Medium	Later, once dinks exist

A reel holds the top-scoring clips in time order, about 5 to 8 clips and 30 to 90 seconds in total. Each clip starts about 1 second before the serve and ends about 1.5 seconds after the rally ends. The reel opens with a title card showing the player's photo crop, the game number and the date, and no score. A small marker under the player's feet, taken from tracking, shows who the reel is about. In the prototype the renderer cuts and joins the clips with FFmpeg into an MP4 for every player. In M4 a web player plays the clips straight from PickleCam's video and draws the marker from the tracking data, and the renderer runs only when a player taps download or share. It re-encodes, because cuts without re-encoding can only start on keyframes.

A reel is good if every clip contains the whole rally, the player appears in every clip and the marker sits on the right person. Beta players also rate their reels. The eval command checks clip boundaries against the rally gold labels, as section 10 describes.

3.4 Accuracy, coverage and review

Requirement	Target	Measured on
Rally winners	The winning team matches gold on at least 95% of rallies, provisionally	Rally-level gold test set, about 30 games plus the 8 full-event test games
Per-player counts	For each stat, off by at most 1 in at least 90% of the player-games where it's shipped, and better than a trivial baseline	Full-event gold test set, 32 player-games
Depth	The bucket matches gold for at least 90% of visible serves and returns, provisionally	Full-event gold
Hits and bounces	F1 no more than 0.03 below human agreement, provisionally	Full-event gold
Highlight clips	At least 95% of clips contain the whole rally, and the marker is on the right player	Rally-level gold and beta review
Partner identity	The server's identity is right at a high rate, set after the M0 spike, for the near and the far team separately	Hand-labelled serves, then the rally-level gold set
Coverage	Reported overall and per camera setup. Provisionally each stat ships for at least 70% of player-games overall. Floors per setup are set after M1	All gold sets

"Off by at most 1" is easy to meet for stats that are often 0 to 3 per player-game, such as winners and errors. A model that always answered 1 could pass. So the eval command reports a trivial baseline next to each stat, which predicts every player's median count from the training games, and a stat ships only if it reaches 90% and beats that baseline.

Coverage depends on the camera. On low down-the-line cameras, near-court bounces are often out of frame, so serve-in and depth for serves into the near court can't reach 70% there. Coverage is therefore reported for each camera setup, and each setup gets its own floor once M1 shows what is possible. The 70% figure stays as a provisional overall target.

The test sets are small. If 27 of 30 games are right, the true rate lies somewhere between about 74% and 97%, so every number comes with a 95% confidence interval, and the test sets grow over time. Below a confidence threshold a stat is marked as not shipped, with a reason such as "near-court bounces aren't visible from this camera". Beta-review corrections set the thresholds. During the beta an owner reviews every game, which takes about 10 minutes, and the corrections become training data.

3.5 Out of scope, and constraints

The v1 has no PickleCam integration, email sending, accounts or web app. It has no score, no singles, ball speed, spin, forced versus unforced errors, forehand versus backhand or dink stats. It doesn't use audio, and it flags close line calls but never rules on them.

Constraint	Consequence
Two new developers who build with Claude Code	Small tested modules, clear contracts, PRs with eval numbers, docs written for agents
Spend as little as possible, with no contractors	Cheap cloud GPUs with a $50 monthly development budget, free tools, R2 storage at about $1 a month, we label ourselves, and production targets under $0.01 a video
A public repo	AGPL-3.0, and no secrets, footage, facility names, session ids or emails in it
MacBook Airs for development, cloud GPUs for compute	The code runs on macOS and Linux, all GPU work uses one Linux Docker image, and CI tests Ubuntu and macOS
Any Bunny key can delete customer videos	We download the prototype's games from the dashboard, and after the prototype read videos only through a read-only endpoint that PickleCam builds

4. Glossary

4.1 Development

Term	Meaning
Repo, branch, PR	The project folder tracked by git, a separate line of work, and a request to merge a branch into `main` that the other owner reviews
CI	Checks GitHub runs on every PR: lint, types and tests
Lint, type checker	ruff catches mistakes and enforces style. pyright checks that values are used consistently
Unit test, smoke test	A unit test checks one function. A smoke test runs a tiny example end to end to check nothing crashes
Spike	A short experiment to learn whether an approach works
Gate	A measurable condition to meet before moving on
CLI	A command-line tool. Ours is `dink`, as in `dink eval --split test`
uv, extra, wheel	uv installs Python and libraries. An extra is an optional group of libraries, such as `cv` with PyTorch. A wheel is a pre-built package
API, endpoint	A way for programs to talk over the web. An endpoint is one URL that answers one kind of request
Contract, schema, pydantic	A contract is the exact format of a file passed between stages. A schema describes it. pydantic defines formats as Python classes and validates data

Term	Meaning
JSON, Parquet	A text format for structured data, and a compact format for large tables
R2, rclone	Cloudflare's file storage, and a tool that syncs files with it
`.env`	A local file of secrets that is never committed
HMAC	A one-way code made from text and a secret key. Nobody without the key can recompute or reverse it
ADR	A short record of one decision and its reasons, kept in `docs/adr/`
Docker image, pod	A packaged copy of our code and its libraries that runs the same on any machine, and a rented remote Linux machine with a GPU
Serverless GPU	A cloud GPU billed per second only while a job runs, then shut down
Spot instance	A cheap cloud machine the provider can take back at short notice. Fine for jobs that can retry
Pull zone, volume network	A Bunny configuration that serves files, and Bunny's cheaper delivery network with fewer locations
Agent, skill	Claude Code working from a written issue. A skill is a packaged workflow it follows, such as `/implement`
Licenses	MIT, BSD and Apache allow any use if you keep the notice. LGPL allows use as a library. AGPL-3.0 requires anyone who distributes the code, or runs a changed version as a network service, to offer its source to users. A CLA lets owners relicense contributions

4.2 Video and computer vision

Term	Meaning
Frame, fps, timestamp	One image, frames per second, and the exact time a frame is shown. With dropped frames, frame 900 isn't at 30.0 seconds
Codec, keyframe, decode	H.264 compresses video. A keyframe is stored whole, and the frames after it store changes. Decoding turns video back into images
HLS, remux, FFmpeg, ffprobe	A streaming format made of small chunks, repackaging video without re-compressing it, and the standard video tools. ffprobe reads timestamps
NVDEC, NVENC, TensorRT	NVIDIA's video decoder and encoder chips on the GPU, and NVIDIA's tool that makes models run faster
Detection, bounding box	Finding an object, usually as a rectangle with a confidence
Tracking, tracklet	Linking detections of the same object across frames. A tracklet is a short unbroken piece of a track
ByteTrack, ReID	A standard tracking algorithm based on motion, and re-identification of a person by appearance
Pose, keypoints	Body joints such as wrists, hips and ankles, found as points
RF-DETR, RTMPose, COCO	A person detector and a keypoint model, both Apache-2.0, and the public dataset most detectors learn from
Heat map	A grid of scores for how likely the ball is at each pixel. The peak is the prediction
Background image	The empty court, made from the median of many frames, so moving things vanish
Calibration, intrinsics, extrinsics	Working out the lens (focal length, centre, distortion) and the camera's position and angle, so pixels convert to court feet
Pinhole model, lens distortion	The standard maths of how a camera projects points, and the way wide lenses bend straight lines
Homography	A mapping between the floor and the image. It is only valid for points on the floor, such as feet and bounces
Reprojection error	The distance in pixels between known court points and where the calibration puts them
Occlusion, visibility mask	One thing hiding another, and the parts of the court a camera can see

4.3 Machine learning

Term	Meaning
Model, training, inference	Behaviour learned from examples, the learning, and running the model on new data
Label, checkpoint	A correct answer written by a person, and a saved trained model
Pretrained, fine-tuning	Already trained on general data, and trained further on ours
Train, dev and test splits	Data to learn from, data to tune choices on, and data used only to report results
Held-out facility, overfitting	A facility used only for testing, and a model that memorises its training data
Gold, silver labels	The hand-made answer key, which never trains, and model output corrected by a person, which does
Self-training	A model labels new data and retrains on its confident labels
Confidence	The model's certainty from 0 to 1. It decides review order and which stats ship
Precision, recall, F1	Of what we found, the share that was right. Of what was there, the share we found. F1 combines both
Tolerance	How close in time a detection must be to count, such as 2 frames
Event spotting	Finding the exact frame of a hit or bounce
HMM, Viterbi	A model of hidden states, such as which candidates are real events, seen through noisy evidence, and the algorithm that finds the most likely sequence of states
Encoder-decoder, U-Net	A network that shrinks an image to understand it, then grows it back to locate things
Loss, weighted cross-entropy	The number training reduces, and a version that weights rare things like the ball more
Augmentation	Extra training examples made by changing brightness, adding blur or flipping images
Gradient-boosted trees	A classic method for tables of numbers that works with little data
Robust loss	A fitting method that limits how much one bad click can move the result
MAE, confidence interval	The average size of the errors, and the range likely to hold the true value
GPU, CUDA, MPS	A graphics card for fast ML, NVIDIA's software for it, and Apple's equivalent

4.4 Pickleball and project terms

Term	Meaning
Rally, side-out	Play from the serve until the ball is dead, and the moment the serving team loses the serve
Side-out scoring	Only the serving team scores. Each team gets two servers per turn, except the first turn of a game, which starts at `0–0–2` . Games go to 11, win by 2. Players often get it wrong, so DinkData doesn't report the score
Rally scoring	Either team can score on any rally. Supported, because no stat reads the score
Starting positions	A team's players stand on their starting sides whenever that team's score is even, and switch only after winning a point on their own serve. Each service turn starts from the right. DinkData uses this only as an internal tiebreak
Two-bounce rule	The serve and the return must bounce before being hit. Volleys are allowed after that
Kitchen, service court	The 7-ft no-volley zone by the net, and the diagonal box a serve must land in
Fault	A rule break that ends the rally, such as a serve out, a volleyed return, a kitchen volley, the ball hitting a player or a foot fault
Volley, dink, lob, stacking	A shot before the bounce, a soft shot into the kitchen, a high shot, and partners lining up on non-standard sides
Net-cord serve, close call	A serve that clips the net and lands in is live. A close call is a bounce near a line that we flag for review
Catalog	Our list of known videos, with no personal data

Term	Meaning
Serve state	Who serves, which partner, from which side, and who receives
Game mode	Whether the players are in a game, warming up, drilling or idle
Rally grammar	The rules of what can follow what inside a rally
Studio	Our web app for labelling, calibration and review
Patch	A human correction stored on top of model output
Coverage	The share of stats we ship
Eval	Our command that scores pipeline output against the gold set

5. System architecture

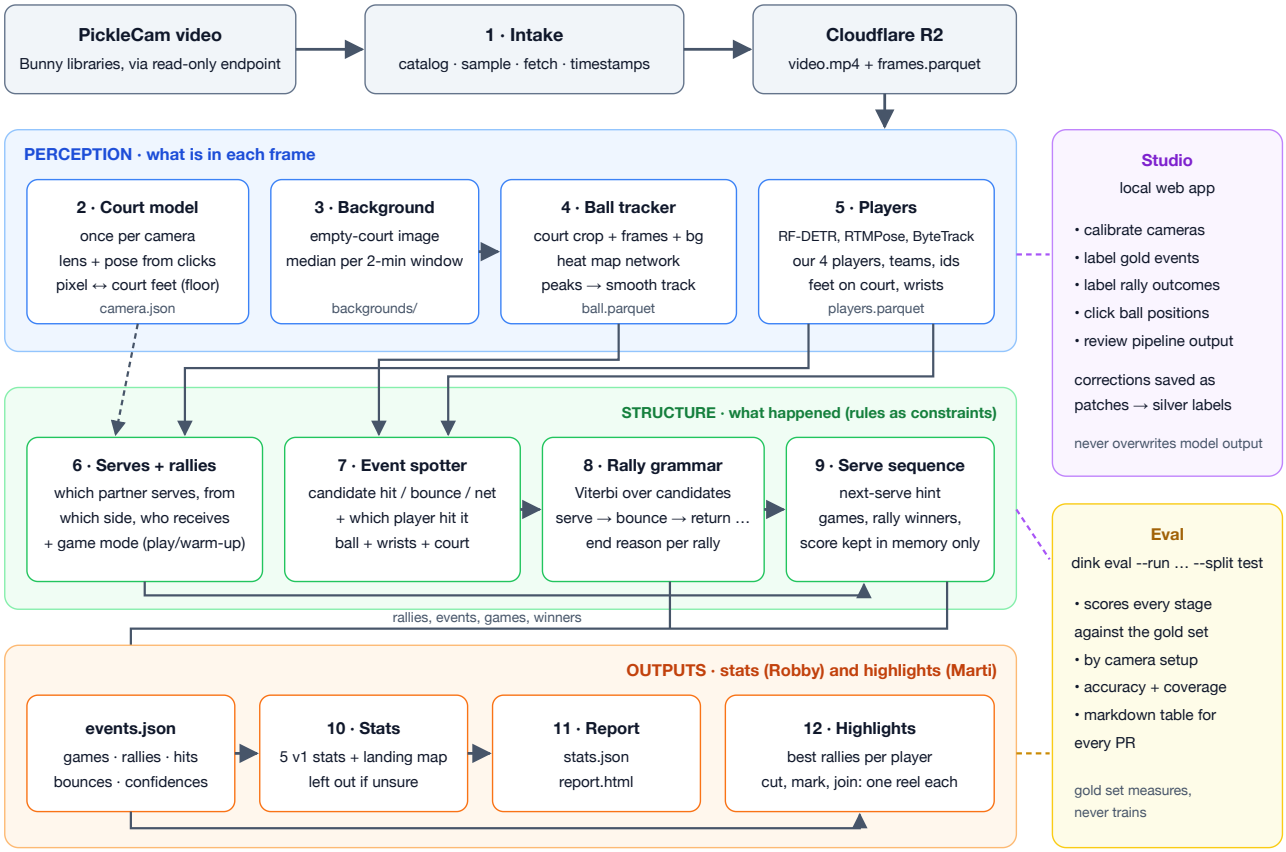


Figure 1. From video to stats and reels. Blue is perception, green is structure and orange is outputs. The box numbers match the table below.

Each stage reads and writes files in fixed formats called contracts, defined in section 7. A stage can therefore be built and tested alone, using the saved outputs of earlier stages.

#	Stage	Output	Owner
1	Intake: catalog, sample, fetch	Catalog, videos without audio, frame timestamps	Robby
2	Court model, once per camera	`camera.json` : lens, pose, court mapping, visibility mask	Robby
3	Background	Empty-court images, one per 2-minute window	Robby
4	Ball tracker	`ball.parquet` : ball position, visibility and confidence per frame	Robby
5	Players	`players.parquet` : 4 players with boxes, keypoints and feet on the court, plus photo crops	Marti

#	Stage	Output	Owner
6	Serve states, game mode, rallies	The `serve` block and boundaries of each rally, play and warm-up segments	Marti
7	Event spotter	Candidate hits, bounces and net events with probabilities	Robby
8	Rally grammar decoder	Clean rallies with events and end reasons	Robby
9	Serve sequence	Games and each rally's winner. Any score stays in memory	Marti
10	Stats	`stats.json`	Robby
11	Report	`report.html`	Robby
12	Highlights	`highlights.json` with a clip list per player, and a reel MP4 per player	Marti
	Studio and eval	Labels, calibrations, patches, metrics	Both

This split gives each owner the stages behind their priority. Robby owns the ball and events path that the stats need. Marti owns players, rallies and the serve sequence, which the highlights need first. It replaces the earlier split, in which Marti owned stats.

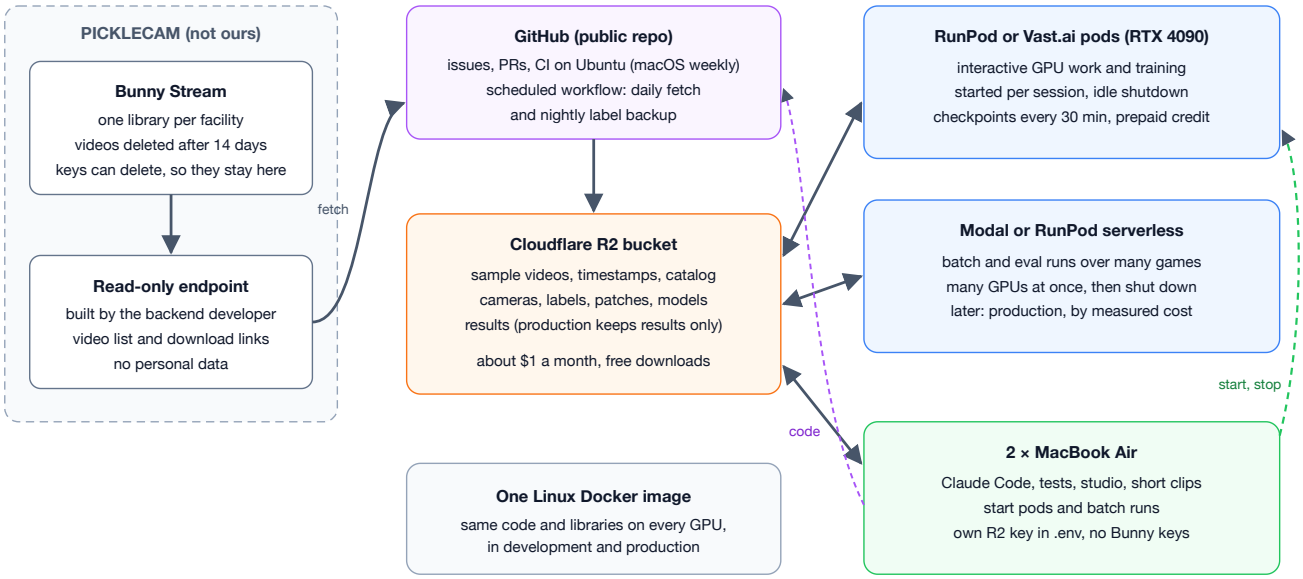


Figure 2. Videos flow from PickleCam through a read-only endpoint into our R2 bucket. Until the prototype is approved, we download them from the dashboard by hand instead (ADR 0018). Rented cloud GPUs do all the GPU work, in development and production. Section 11 covers the costs.

The architecture follows these rules.

- Every file passed between stages has a contract, and a stage that writes an invalid file fails.
- Time always comes from timestamps.
- The court mapping applies only to points on the floor.
- Every detection carries a confidence.
- A rule-based version of each stage comes before its model.
- Every change to a model, threshold or rule shows eval numbers before and after.
- Human corrections are stored as patches and never overwrite model output.
- No personal data appears in code, labels, logs or PRs.

6. The pipeline, stage by stage

6.1 Intake (`dinkdata.data`, Robby)

For the prototype, intake starts from games downloaded by hand (ADR 0018). The owners pick about 150 games from about 40 facilities. They hold out whole facilities for testing, at least one per camera setup and more for low down-the-line cameras. They spread the rest across cameras, court colours and game lengths, and skip recordings under 5 minutes. The split never changes once labels exist. Robby downloads each game's 1080p MP4 from its watch page within 14 days of upload.

`dink ingest` takes one downloaded MP4. It drops the audio track with `-map 0:v`, saves every frame's timestamp with ffprobe, and flags dropped and duplicated frames. It asks for the session id, facility and court at a hidden prompt, gives the game its HMAC `video_id` and `camera_id` (ADR 0016), adds a catalog row and uploads the results to R2, skipping files already there. `dink run <video_id>` finds the game's camera, and so its calibration, in that row. Running on a loose MP4 and recognising its camera automatically wait for M4. Bunny's MP4 keeps dropped-frame gaps, and the duplicate check stays in case a camera or encoder fills them.

After the prototype, intake runs on its own. `dink catalog sync` reads PickleCam's read-only endpoint, specified in section 8.2, and writes one row per video. Each row has an opaque `video_id`, which is an HMAC of the session id keyed with a secret in `.env`, plus hashed facility and camera ids, the upload time, length, frame rate, size, the camera setup once calibrated, and the split. The map from our ids back to session ids lives only in a private `catalog/id_map.parquet` in R2.

`dink sample` applies the same selection rules to the catalog.

`dink fetch` downloads each new game's 1080p MP4. Without one, it takes the original upload, and otherwise the HLS stream, which it remuxes with `ffmpeg -c copy` without quality loss (ADR 0017), and passes the file to the same steps as `dink ingest`. A scheduled GitHub Actions workflow runs fetch daily, so selected games are copied before PickleCam deletes them.

In production, fetch doesn't copy videos into R2. Catalog sync puts each new video on a job queue, and a GPU worker reads the video from Bunny, processes it and stores only the results (section 11.5).

6.2 Video I/O (`dinkdata.video`, Robby)

PyAV, which includes FFmpeg on every OS, reads frames with their timestamps, both in sequence and by exact random access. Every duration and speed uses timestamps. A pre-commit check and code review catch `frame / 30`, because dropped frames make it drift by seconds over a long game.

6.3 Court model and calibration (`dinkdata.court`, Robby)

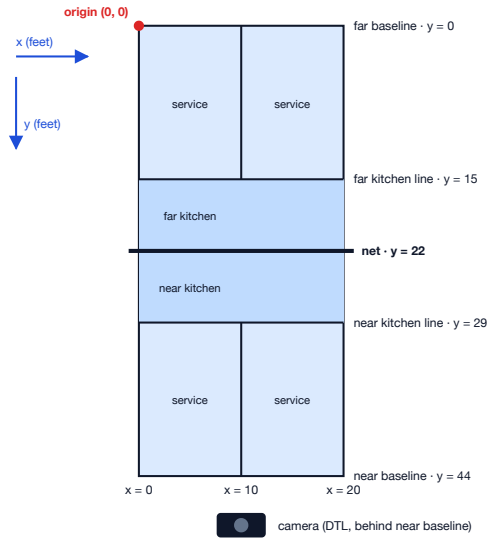


Figure 3. Court coordinates in feet. The origin is the far-left baseline corner as seen from the camera. x runs 0 to 20 across the court and y runs 0 to 44 toward the camera. The net is at y = 22 and the kitchen lines are at y = 15 and y = 29.

1. A person clicks the visible named court points, several points along each visible line, and the top of the net, which is 36 in high at the sidelines and 34 in at the centre. A command-line click tool comes first and a studio mode later.
2. A solver fits a pinhole camera with radial lens distortion, using `scipy.optimize.least_squares` with a robust loss. It moves points to their places, line points onto their lines and the net tops to their heights. Rec nets can sag, so the net heights are soft constraints.
3. One partial court can't pin down a lens, so the solver fits lens parameters jointly across cameras of the same model. Each camera then gets its own pose and a small distortion correction.
4. The reprojection error on held-back clicks decides whether a calibration passes. The first 20 calibrations set the threshold.
5. The visibility mask is the image border projected onto the floor, clipped at the horizon. A bounce outside it is not visible, rather than out.
6. `image_to_court` and `court_to_image` convert between pixels and feet, with tests on synthetic cameras.

The prototype needs this one-time calibration for each camera before `dink run` works on its games, including the held-out facilities' cameras, since calibrating isn't training.

From M1, each video's background is compared with the projected court lines. A mismatch flags a camera that moved. A calibration takes about 2 minutes, so our sample of about 100 cameras takes a few hours. All of PickleCam could reach 1,400 cameras, about 47 hours of clicking, so M4 trains an automatic calibration model on our own calibrations.

6.4 Background (`dinkdata.ball`, Robby)

The background step samples one frame per second and takes the pixel median of each 2-minute window. It stores the images in `runs/{run_id}/{video_id}/backgrounds/`. Separate windows handle changes in lighting and camera exposure.

6.5 Ball tracker (`dinkdata.ball`, Robby)

1. The tracker crops the frame to the court, with room above for lobs and margin outside the lines for out balls. The far ball keeps more pixels than if the whole frame were shrunk. It masks the overlays.
2. The input is 3 to 8 consecutive crops plus the background crop, at 288×512 to start.
3. A TrackNet-style encoder-decoder outputs a heat map per frame. It trains with weighted cross-entropy on small Gaussian targets. We write it from scratch, so the architecture can change later.
4. Heat map peaks link into smooth tracklets. Tracklets that never cross our net, or bounce outside our court, belong to neighbouring courts and are dropped. Gaps of up to 10 frames are filled with a fitted curve and marked `interpolated`.
5. Training starts with about 4,000 hand-clicked frames in 2-second clips from about 30 training games. After that, people accept or fix the model's uncertain frames. About 1,000 separate test frames measure it. Augmentation changes brightness, adds blur, shifts crops and flips left to right.

This is the best recipe in the 2025 benchmark, and fixed cameras suit it. The WASB comparison and self-training wait until v1 stops improving. On low cameras the far ball is tiny, which the crop, larger inputs and more labels address. At a hit the ball is often hidden, so the event spotter also uses wrist keypoints and heat map peaks.

6.6 Players, serve states, game mode and rallies (`dinkdata.players`, Marti)

1. RF-DETR finds people in the court crop, and RTMPose finds their body keypoints. Both are pretrained, so they need no labelling, and both are Apache-2.0. Each sits behind a small interface.
2. ByteTrack, through the MIT-licensed `supervision` library, links detections into tracks. ByteTrack uses motion only, so an appearance model, such as OSNet from torchreid (MIT), re-links people after occlusion and tells partners apart.
3. A player's foot point is the midpoint of the ankles, or the bottom of the box. If the box touches the bottom of the frame, the foot point comes from the hips and gets a lower confidence. The floor mapping then places it on the court.

4. The four tracks that spend the most play time on our court, with generous margins behind the baselines, become the players. Fragments of one person merge by side and appearance.
5. Ids follow one convention in the pipeline and in gold labels. p1 is the first server in the recording and p2 is p1's partner. p3 receives the first serve and p4 is p3's partner. Team A is p1 and p2, and team B is p3 and p4. Appearance carries ids across games and end switches. Keeping partners apart matters most, so each serve checks the server against both stored profiles.
6. **Partner identity is the riskiest assumption in the plan.** Per-player serve stats and the next-serve winner hint depend on telling p1 from p2 at every serve. Partners often dress alike, the far team is small on low down-the-line cameras (64% of videos), and the near team is seen from behind. The player spike in M0 therefore measures partner-identity accuracy at the serve, separately for the near and far team, on serves the owners label by hand. The eval command keeps reporting it. M2's design is locked only after the spike shows the accuracy we can reach. If the far team falls short, per-player serve stats ship only for the camera setups that pass, and rally winners lean on the ball from M3.
7. At each serve formation the serve state records which team serves, which player, from which side and who receives. Internally it also notes whether each team stands on its starting sides, which hints at score parity. Players who lose track of the score and teams that stack both make that hint wrong, so it only breaks a tie when the appearance check can't tell partners apart, and it never goes into a file.
8. Rules based on serve formations and their rhythm label segments as `game`, `warmup`, `drill` or `idle`.
9. A rally starts at a detected serve and ends when the ball is dead: the ball leaves play, players relax and the next serve formation appears. In M2 rallies simply run from serve to serve.

The main risks are occlusion near the net, spectators near the court, end switches and partners swapping ids. The appearance check catches swaps, with the parity hint as a tiebreak.

6.7 Event spotter (`dinkdata.events`, Robby)

1. The spotter proposes candidates and aims to miss fewer than 1% of real events. A candidate is a sharp change of direction, a V-shaped dip in image height, which is typical of a bounce, a jump in speed, the start or end of a track, or a wrist close to the ball.
2. For each candidate it computes features over 10 frames on each side. These are ball motion, court position where a floor contact is plausible, each player's wrist distance and wrist speed, the side of the net, and heat map strength.
3. Version 0 uses rules, so a change of direction near a swinging wrist is a hit. Version 1 uses gradient-boosted trees, and version 2, if needed, a small temporal network. The output is the probability of a hit, bounce, net event or nothing, and for hits the probability of each player.
4. Only silver labels train it. In M3, people correct the rule-based version's suggestions on about 10 training games, and beta corrections add more later. Gold games never train.

6.8 Rally grammar decoder (`dinkdata.events`, Robby)

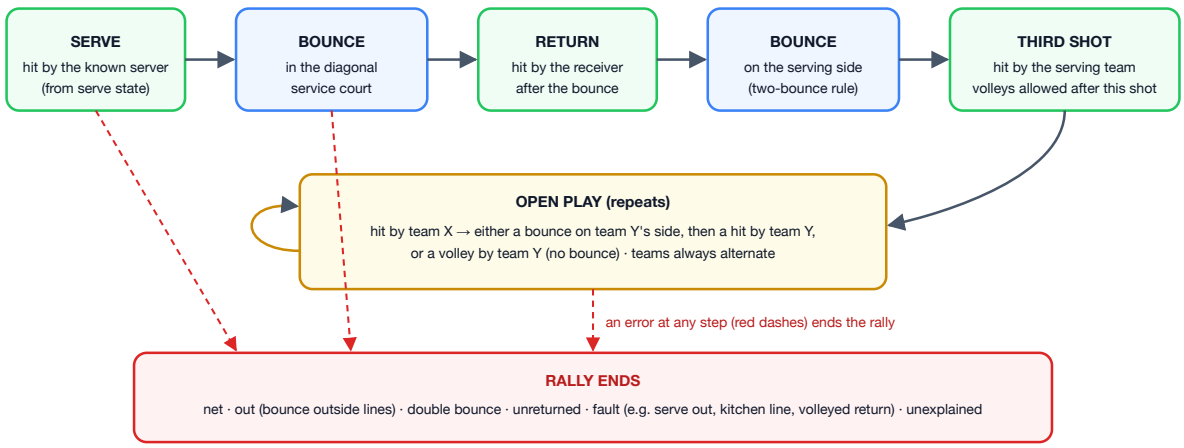


Figure 4. The rally grammar, simplified. The dashed exits end the rally.

The serve, by the known server, must bounce in the diagonal service court. The receiver returns it after the bounce, and the return must bounce on the serving side. After that, hits alternate between teams, each followed by a bounce on the other side or a volley by the other team. A rally ends as `net`, `out`, `double_bounce`, `unreturned`, `fault` or `unexplained`.

Viterbi goes through the candidates in time order and accepts each as an event, with its hitter, or rejects it. The classifier's probabilities and the grammar's transitions score each choice. When the rules require an event the spotter missed, the decoder can insert a low-confidence hit. Some legal but unusual sequences are allowed with a penalty, so the decoder never forces nonsense. These are the ball hitting a player, volleyed serves or returns, kitchen volleys, replays, players stopping on a wrong call and net-cord serves.

The output is a clean event list, the end reason and the winner implied by the ball. When the ball-based winner's confidence passes its threshold, it decides the rally. Otherwise the next-serve hint decides. A disagreement lowers the rally's `conf`, so a stat can leave the rally out instead of guessing. Out calls follow what the players did.

6.9 Serve sequence (`dinkdata.serves`, Marti)

The serve sequence turns serve states into each rally's winner and the game boundaries. DinkData never reports the score (ADR 0019), and no stage depends on it, because PickleCam players often lose track of side-out scoring.

1. **The next-serve hint.** Each serve is compared with the next one only. If the other team serves next, the serving team lost. If the same player serves again from the other side, the serving team won. If the partner serves, the rules say the team lost with server 1, but a team that won and forgot to swap looks the same. So the serves can't tell, the ball decides from M3, and until then the rally gets a low confidence. A scoring mistake by the players changes at most one rally and doesn't carry through the game.
2. **The internal side-out replay.** The rules of side-out scoring, run over the whole game, add weak evidence where the next-serve hint can't tell. The replay lives in memory only. It allows rule breaks with a small penalty, drops out when the serves don't fit the rules, and writes nothing to `events.json`. Its rules are below.

Previous rally	Next serve under the rules
Serving team won	The same server serves from the other side, because the partners swapped
Serving team lost with server 1	The partner serves as server 2, from where they stand
Serving team lost with server 2, or the first server at `0–0–2`	Side-out. The other team serves from the right
Replay	Same server, same side, same score

3. **Games.** A game boundary comes from a break in play: the players stop, walk, swap ends or reset their formation for a first serve from the right. The replay's guess at a game end is only a hint. A game has an id, a start, an end and its rallies, and no target, winner or final score.
4. If fewer than four players are on court, the recording is singles, which v1 doesn't support. Any scoring system works, because no stat reads the score.

6.10 When players break the rules

Recreational players make mistakes. They forget who served last, the wrong partner serves, a team serves again after it should have lost the serve, a player serves from the wrong side, or they lose track of the score and end the game at the wrong number. Most PickleCam games have a few of these. DinkData treats this as normal, in three ways.

1. **No output depends on the score.** The players know their own score, so a score that differs from theirs would look wrong at once, even when our count matches what was played. No report, stat or reel shows the score, games won, points scored, rallies won or rule deviations.
2. **Rally winners come from local evidence.** The next-serve hint looks only at the next serve, and the ball decides when it is confident, so one mistake by the players changes at most one rally.
3. **Gold labels record what was played.** The rally labeller replays the score and flags rallies where the serves break the rules, and the labeller marks each rule deviation in `rallies.json`. The labeller marks every rally's winner by hand, so the replay checks the winner instead of choosing it. The rally-level gold set therefore shows how often players get the score wrong, and that number never reaches a player.

6.11 Stats and report (`dinkdata.stats`, `dinkdata.report`, Robby)

Each stat is a pure function that returns a value, numerator, denominator, confidence, shipped flag and reason. Unit tests cover every rule. A serve on the kitchen line is a fault, a winner requires that no opponent touched the ball, and a serve into an area the camera can't see is left out. The report is one static HTML file built from a Jinja2 template, laid out for phones, with an inline SVG landing map. It shows no score and no rule deviations.

```
{"video_id": "v_9c1e…", "game_id": "g1",
 "players": {"p1": {
   "serve_in_pct": {"value": 0.875, "num": 7, "den": 8, "conf": 0.93, "shipped": true},
   "winners": {"value": 1, "conf": 0.64, "shipped": false, "reason": "low_attribution_conf"}}}}
```

6.12 Highlights (`dinkdata.highlights`, Marti)

1. The selector scores every rally for each player with the rules in section 3.3 and keeps the top clips within the length limit.
2. Clip edges come from rally boundaries with padding. They never cut into the next serve.
3. It writes `highlights.json` with each clip's start, end, reason and score, so a person can check or change the selection in review.
4. The renderer writes a reel MP4 for every player: FFmpeg cuts the clips, burns in the marker and title card, and joins them.
5. In M4, a web player plays each reel from PickleCam's video with the title card and marker drawn on top, and a serverless job renders the MP4 only on request, encoding with NVIDIA's NVENC, and keeps it for 30 days (ADR 0012).

In M0 the player and renderer work from a hand-written list of rallies, so they need no models. In M2 it uses detected rallies and their winners. In M3 it adds winners and aces.

6.13 Studio (`dinkdata.studio`, both)

The studio is a local web app for every human task. A FastAPI server decodes frames by exact timestamp, because browsers can't land on an exact frame, reads and writes R2, and validates every save against the contracts. A small TypeScript page shows the frame, overlays and a timeline.

Mode	Milestone	Keys
Events	M1	Left and right arrows move one frame, with Shift for 10. R marks rally start or end. H then 1 to 4 marks a hit and its hitter. B then a click marks a bounce. X marks a bounce out of view. N marks the net. W then A or B marks the winner. Enter accepts a suggestion, Delete rejects it and J jumps to the next uncertain item
Rally level	M0, as a minimal tool	A single local web page with a video player. Mark each serve's time, server and side, and each rally's winner. It needs no exact frames, so a browser player is enough. The studio absorbs it later and proposes serves from M2
Calibrate	M1, after the click tool	Click points, lines and net tops, then check the fitted court and its error
Ball	CVAT first	Ball clicks start in CVAT in M0. A converter turns CVAT's export into our `ball_labels` contract and has a test that checks frame alignment, because misaligned frames broke the prototype's labels. Once a ball model exists, CVAT can show its predictions for people to accept or fix. A studio ball mode is optional later
Review	M4	Items appear lowest confidence first, including highlight selections. Edits are saved as patches

After H or W, the next key names a player or team, so the keys don't clash. The studio writes our contract directly, its pickleball shortcuts are faster than a generic tool's, and it becomes the review tool. Labelling is the critical path, though, so no label set waits for the full studio. Rally-level labels start in M0 with the minimal tool, ball clicks start in M0 in CVAT, and calibration uses the click tool. Only full-event gold waits for the studio's events mode in M1.

6.14 The feedback loop

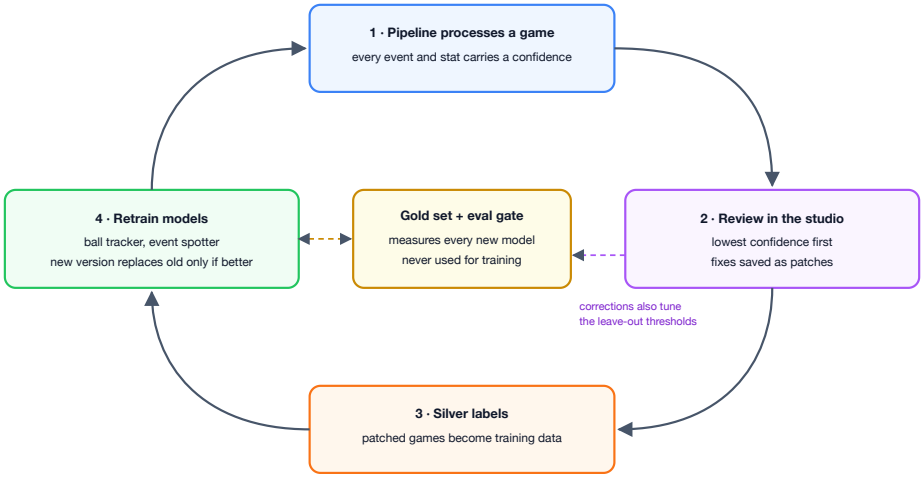


Figure 5. Review corrections become silver labels, which retrain the models. A new model replaces the old one only if it scores better on the gold set, which never trains.

7. Data contracts

Contracts are pydantic models in `src/dinkdata/contracts/`. The JSON Schemas in `contracts/` are generated from them, and CI fails if the committed schemas differ from freshly generated ones. Parquet tables have pyarrow schemas. Every stage validates what it reads and writes at run time. Each file has a versioned `schema` name, and a breaking change needs both owners. Every detection has `conf`. Times are given as `frame`, for lookups, and `t`, in seconds from the first frame's timestamp. Files written in a browser, `rallies.json` and the clip times in `highlights.json`, store `t` and leave `frame` null until a later step fills it. Court positions are in feet, and `null` means not visible. No contract holds personal data.

File	Contents
`catalog.parquet`	`video_id`, `facility_id`, `camera_id`, `uploaded_at`, `duration_s`, `fps_nominal`, `size_bytes`, `setup`, `split`, `fetched`, `notes`. `uploaded_at` can be null for a hand download and `setup` stays null until calibration. Every game of a facility has the same split
`{video_id}.frames.parquet`	`frame`, `t`, `dropped_before`, `duplicate`. One row per decoded frame, and the first frame's `t` is 0
`camera.json`	Camera id and model, setup, image size, intrinsics, distortion (k1, k2), rotation, translation, reprojection error, visible region in court feet, overlay masks in pixels, source, version
`ball.parquet`	`frame`, `t`, `x`, `y`, `visible`, `conf`, `interpolated`, the top 3 `peaks`
`players.parquet`	One row per frame per track: `frame`, `t`, `player_id`, `track_id`, `team`, `box`, 17 `keypoints`, `foot_court`, `estimated_foot`, `conf`, `foot_conf`. Tracks that aren't players have a null `player_id`
`candidates.parquet`	For each candidate, `frame`, `t`, `conf`, the hit, bounce and net probabilities, and the hitter probabilities for p1 to p4
`events.json`	The game log, shown below
`stats.json`	For each game, subject (player or team) and stat: value, num, den, conf, shipped and reason. Depth stats hold bucket counts and the landing map holds court points
`highlights.json`	One reel per game per player: the clips with rally, start, end, reasons and score, and the MP4 once rendered
`rallies.json`	Rally-level gold labels: games, and for each rally the server, side, receiver, winner, score as played and any rule deviation. Only gold keeps the score, to measure how often players get it wrong. Times are `t` only, because the browser labeller can't know frames
`ball_labels.parquet`	Ball clicks: `frame`, `t`, `x`, `y`, `visible`, `clicked`, `split`

File	Contents
`patch.json`	Corrections to one pipeline run, with its `run_id`. Add, change and delete operations on games, rallies and events by id, with author handle, time and the old values. The patched `events.json` is saved next to it


```
{"schema": "dinkdata.events/v2",
 "video": {"id": "v_9c1e…", "fps_nominal": 30, "frames": "videos/v_9c1e….frames.parquet",
"dropped_frames": 14},
 "camera": {"id": "c_41ab…", "version": 2},
 "players": [{"id": "p1", "team": "A", "label": "Player 1", "crop": "crops/p1.jpg"}],
 "segments": [{"type": "warmup", "start_frame": 0, "end_frame": 5400, "start_t": 0.0, "end_t":
180.43, "conf": 0.9}],
 "games": [{"id": "g1", "start_frame": 5520, "end_frame": 27360, "start_t": 184.47, "end_t":
912.30,
   "rallies": [{"id": "g1_r1", "start_frame": 5520, "end_frame": 5910, "start_t": 184.47,
"end_t": 197.47,
     "serve": {"server": "p1", "side": "right", "receiver": "p3", "conf": 0.96},
     "winner_team": "A", "winner_source": "next_serve", "replay": false,
     "end_reason": null, "reviewed": false, "conf": 0.94,
     "events": [
       {"id": "e1", "type": "hit", "frame": 5531, "t": 184.83, "player": "p1", "shot_index": 1,
"volley": false, "conf": 0.97},
       {"id": "e2", "type": "bounce", "frame": 5562, "t": 185.87, "image_xy": [880.5, 412.0],
        "court_xy": [6.1, 9.4], "in": true, "close_call": false, "conf": 0.95},
       {"id": "e3", "type": "hit", "frame": 5575, "t": 186.30, "player": "p3", "shot_index": 2,
"volley": false, "conf": 0.91},
       {"id": "e4", "type": "bounce", "frame": 5609, "t": 187.43, "image_xy": null,
        "court_xy": null, "in": null, "close_call": false, "conf": 0.60}]}]}]}
```


The example shows four things. The return's bounce, e4, lands below the frame, so its position and `in` are `null`. The ball doesn't show how the rally ended, so `end_reason` is `null`. The next serve, p1 again from the left, shows that team A won, so `winner_source` is `next_serve`. And `t` differs from `frame / 30` because 14 frames were dropped earlier.

The end reasons are these.

- `net`: the shot didn't cross the net.
- `out`: the first bounce was outside the lines.
- `double_bounce`: the ball bounced twice on one side.
- `unreturned`: a player touched the ball but didn't send it back over.
- `fault`: a rule break such as a volleyed return, a kitchen volley, the ball hitting a player or a foot fault.
- `unexplained`: the rally ended for a reason we can't see.
- `null`: unknown.

8. Data, storage and privacy

8.1 Storage

All files live in one Cloudflare R2 bucket named `dinkdata`. R2 has no object versioning, so labels, calibrations and patches are written once to versioned paths and never overwritten. A nightly scheduled job copies `labels/` and `cameras/` to a second bucket as a backup, which costs cents because those files are small.

```
catalog/catalog.parquet, catalog/id_map.parquet (private)
videos/{video_id}.mp4, videos/{video_id}.frames.parquet
cameras/{camera_id}/v{n}/camera.json
labels/gold-events/{video_id}/{labeller}/v{n}.json
labels/gold-rallies/{video_id}/{labeller}/v{n}.json
labels/ball/{video_id}/{labeller}/v{n}.parquet, labels/feet/{video_id}/{labeller}/v{n}.parquet
labels/silver/{video_id}/{run_id}/patch.json and events.json
runs/{run_id}/{video_id}/  backgrounds, ball, players, candidates, events, stats, report,
highlights
models/{name}/{version}/   checkpoint, config, eval results
eval/{run_id}.csv
```

The 150 games take about 60 GB, and everything else a few GB more. R2 charges about $0.015 per GB-month after 10 GB free, with free downloads, so roughly $1 a month. These prices come from comparisons published in mid-2026, so check them at sign-up. Each machine has its own R2 key in `.env`, limited to this bucket. The Macs need write access because the studio saves labels. After the prototype, only the fetch machine holds the endpoint token. Claude Code may not read `.env`.

During M0 to M3, R2 is the shared development store for the games, labels, calibrations and models. `dink run` itself never reads R2: it reads models from a local folder, and a separate command downloads the latest models from R2 once. After the prototype, R2 holds results, not videos. The JSON outputs and report take about 0.5 MB a video, per-frame tracks are deleted after 30 days, and reel MP4s rendered on request are deleted after 30 days. Full videos stay in R2 only for the gold sets and training data.

We chose R2 over keeping videos on Bunny. Keeping them on Bunny would mean changing PickleCam's production cleanup job, mixing research data with customer data, paying for downloads and still needing a second store for labels and models.

8.2 The read-only endpoint, a spec for the PickleCam backend developer

PickleCam builds this after the prototype (ADR 0018). Until then we download games from the dashboard by hand.

Bunny Stream has one key per library, and every key can also delete videos. No Bunny key leaves PickleCam. Instead, the backend developer adds one read-only call protected by a token (ADR 0017).

Call	Returns
`GET /dinkdata/recordings?since=&page=`	A paged list across facilities with `session_id`, `facility_id`, `camera_id`, `started_at`, `stopped_at`, `uploaded_at`, `expires_at`, `library_id`, `video_id` and `mp4_url`, the download link for the 1080p MP4

The call returns no personal data, and the facility dashboard already loads the same fields. PickleCam builds `mp4_url`, so a change to its link format or pull zone needs no change on our side. Frame rate, resolution and size come from the file at fetch time.

8.3 Privacy rules

1. No emails, facility names, session ids or Bunny ids go into the repo, issues, PRs, logs or labels.
2. No footage or frames go into the repo. Test fixtures are synthetic.
3. Fetch drops the audio track, so we store no audio.
4. Player photos and reels stay in `runs/` and go only to the players in that game.
5. DinkData stores no contact details. How beta players receive reports and reels is an open question.
6. Agents work only from issues the two owners wrote.
7. Robby confirmed for PickleCam on 17 October 2026 that all games may be used for this project.

9. Labelling and the gold set

Label set	Purpose	Size
Gold, full events	The answer key for events and per-player stats. Never trains	12 games, 4 dev and 8 test from held-out facilities covering all setups, split 6 each between the owners. They also give rally-level labels
Gold, rally level	The answer key for rally winners, serve states, game boundaries and highlight clip edges. It also records the score as played and rule deviations, to measure how often players get the score wrong	About 50 more games, 20 dev and 30 test, about 10 minutes each
Ball clicks	Training and measuring the ball tracker	About 4,000 training frames and 1,000 test frames, in 2-second clips
Foot clicks	Measuring foot position error	About 200 serve frames from test games
Silver	Training the event spotter	About 10 corrected training games in M3, then beta corrections
Agreement game	Measuring how well two people agree, the realistic ceiling	One of the 4 dev games, labelled by both owners

Labels are files that point into the unchanged video by frame and timestamp. That's why exact frame access matters.

To label a full-event game, open it with `dink studio --label <video_id>` and calibrate its camera first if needed.

1. In a rally pass at 2× speed, press R at each serve contact and again when the ball is dead.
2. In an event pass, step frame by frame. Press H and a player number at each contact, B and a click at each bounce, X if the bounce is out of view, and N for the net.
3. Confirm the proposed end reason and winner of each rally.
4. The studio replays the rules over your rallies and flags any rally where the score doesn't fit. Mark it as a rule deviation if the players really did it, otherwise fix your labels.
5. Save. The file goes to `labels/gold-events/<video_id>/<your name>/v<n>.json`, and nothing is overwritten.

A full-event game realistically takes 3 to 5 hours, so the 12 games take each owner 18 to 30 hours during M1. Start with 2 dev and 2 test games, so the eval command has data early.

A rally-level game takes about 10 minutes in the minimal tool, available from M0. Play the video, press a key at each serve, pick the server and side, pick the winner after each rally, mark game starts, ends and cut-offs, and mark any rule deviation. To calibrate a camera, run `dink studio --calibrate <camera_id>`. Click the visible points, pressing S to skip one, then 3 to 5 points per line and the net tops, then check that the fitted court sits on the painted lines. Set the camera setup and save. Ball clicks happen in CVAT, run on a Mac with Docker or on CVAT's cloud plan. Click the ball centre in each frame, or mark it hidden. CVAT can fill frames between clicks along a smooth flight, which you then check. `dink eval agreement` compares the owners' two labels of the agreement game, and the hit and bounce targets come from it.

10. Evaluation and targets

`dink eval --run <run_id> --split dev` or `--split test` prints a markdown table for the PR, broken down by camera setup, and appends the results to `eval/{run_id}.csv` in R2. A small synthetic version runs in CI.

Stage	Metric
Calibration	Reprojection error in pixels on held-back clicks
Ball	Precision, recall and F1 within 10 px at 1080p, also at 5 and 20 px, and the median error
Players	Partner identity at the serve, for the near and far team separately. Hitter identities that match gold. Foot error in feet on foot clicks
Serve states	Server, side and team accuracy

Stage	Metric
Rallies	Start and end within 0.5 seconds, and missed or extra rallies
Hits, bounces, net	Precision, recall and F1 within 2 frames, about 67 ms
Attribution	Hitter accuracy among matched hits
Bounces	Position error in feet, in/out accuracy and close-call rate
Rally winners	The share of rallies whose winning team matches gold, reported separately for games with and without rule deviations in the gold labels
Stats	The share of player-games off by at most 1 for each stat, the mean error, and the same numbers for the median-count baseline
Highlights	The share of clips containing the whole rally, and the share with the marker on the right player
GPU time	GPU-seconds per video for each stage, as information. The cost-per-video target waits for M4
Coverage	The share of stats shipped, overall and per camera setup

Gate	Requirement	Milestone
Partner identity	Accuracy at the serve measured for near and far teams, and a target agreed before M2 starts	M0 spike
Serve states	Server identity at or above the target set after the M0 spike, for near and far teams	M2
Rally winners	The winning team matches gold on at least 95% of test rallies, provisionally	M2
Rally reels ship	At least 95% of clips contain the whole rally	M2
Ball v1	Recall and precision of at least 90% at 10 px on test clicks	M2 to M3
Ball stats ship	Each stat off by at most 1 in at least 90% of test player-games, better than the median-count baseline, and at or above the coverage floor for each setup	M3
Events	F1 no more than 0.03 below human agreement, provisionally	M3
Beta	A week of reviewed games in which reviewers change fewer than 1 stat in 10	M4
Cost	Measured compute plus bandwidth under $0.01 a video on the chosen provider	M4

Nobody trains or tunes on test data. Every accuracy change includes the table from before and after, with 95% intervals. A change counts as better if the total improves and no camera setup gets worse by more than its interval. Results for a setup with only about 2 test games are a warning, not a verdict.

11. Compute and costs

11.1 Where jobs run

All GPU work runs on rented cloud GPUs from the start, in the same Linux Docker image that production will use. The two Windows PCs aren't part of the plan. They stay available as free spare capacity for a long training run, but nothing depends on them.

Job	Where
Coding, tests, studio, short clips without a GPU	MacBook Airs
Interactive GPU work and training	RunPod pods with a community RTX 4090, or RunPod Secure Cloud when community GPUs are unavailable. A pod is a remote Linux machine with a GPU, started for a session and then stopped
Batch runs over many games, and eval runs	RunPod pods until a ticket needs more. The first batch eval ticket decides on Modal, which spreads the games across many GPUs and then shuts down. Vast.ai waits for the M4 benchmark

Job	Where
Daily fetch of selected videos, after the prototype	A scheduled GitHub Actions workflow, which is free for public repos and needs no GPU
Production	The cheapest provider that passes the accuracy gates and finishes each day's videos, chosen by the M4 benchmark

Spending stays low by choosing the cheapest GPU that gives the same result, and accepting slower training. A model trained on a cheap community RTX 4090 for 20 hours is as good as one trained on an expensive A100 for 4 hours, so we pick the cheap one. Cheap GPUs can be interrupted, so every training run saves a checkpoint at least every 30 minutes and resumes where it stopped. We don't save money on anything that changes quality: input resolution, model size, the amount of labelled data, or the accuracy gates. In production, the provider must meet the same accuracy as in testing, and a cheaper option that misses a gate is rejected.

Guardrails stop surprise bills.

- RunPod runs on prepaid credit with auto-pay off, topped up $25 at a time and at most $50 a month, so spending can't run past the balance. RunPod emails Robby below $5.
- RunPod can't stop an idle pod, so every pod starts with a 4-hour stop timer.
- The development budget is $50 a month. A training run that measurably improves accuracy can go over it if both owners agree first. Accuracy wins over saving money, but nothing is spent without a reason.
- Every eval run records its GPU cost, so the cost of an experiment is visible in the PR.

11.2 Assumptions

PickleCam's owner says PickleCam receives about 4,500 videos a day, and the number will grow as PickleCam reaches more courts in the Philippines. Every figure below is an estimate built on these assumptions, which M2 and M4 replace with measurements.

Assumption	Value
Videos a day	4,500, about 135,000 a month
Average video length	15 minutes. The median is 11 to 14 minutes, and some videos run to the 45-minute cap
Video size	About 0.5 GB at 4.4 Mbps, so 2.2 TB a day and 67 TB a month
GPU time per video, first working version	About 9 GPU-minutes on an L4-class GPU: every frame through every model, CPU decoding, plain PyTorch
GPU time per video, optimized	About 1.5 GPU-minutes. Section 11.5 lists the changes that get there
Cloud prices, from mid-2026 trackers	Modal L4 $0.80 an hour. RunPod serverless L4 about $0.48. AWS g6 L4 spot $0.44 to 0.74. RunPod community RTX 4090 about $0.34. Vast.ai RTX 4090 from $0.14. An RTX 4090 runs these models roughly twice as fast as an L4, which we confirm in M4
Bunny bandwidth to our workers	$0.03 per GB on the standard network in Asia, $0.01 in Europe and North America, and $0.005 on Bunny's volume network
R2 storage	$0.015 per GB a month after 10 GB free, with free downloads

11.3 Costs to start and train

Item	Estimate over 6 to 12 months
Training, 200 to 400 GPU-hours on community RTX 4090s at about $0.20 to $0.34 an hour, with no A100 unless a model needs it	About $40 to $140
Batch and eval runs over the 150 sample games, mostly covered by Modal's free credit	About $0 to $100
Interactive pods, about 6 hours a week each for two people	About $80 to $160
Pod disk for code, caches and checkpoints	About $20 to $60
R2 storage for the 150 sample games, about 75 GB	About $1 a month, rising to about $7 if we keep 500 games

Item	Estimate over 6 to 12 months
Bunny bandwidth to copy the 150 games once	About $2, on PickleCam's Bunny bill, which PickleCam covers
CVAT, GitHub and CI	Free: self-hosted CVAT, a public repo, free Actions minutes

The total from setup to the end of the beta is roughly $150 to $450, or about $15 to $40 a month, inside the $50 monthly budget. Owners' labelling time is the real cost of this phase, and Claude Code subscriptions are an existing expense outside this estimate.

11.4 Production cost at 4,500 videos a day

This section matters after the prototype. PickleCam asked us not to spend effort on cost before the prototype is good (ADR 0018). A pipeline built the straightforward way would be expensive.

Line	Straightforward pipeline	Optimized pipeline (this plan)
GPU compute	9 GPU-minutes a video: about $9,700 a month on RunPod serverless L4s, or $16,200 on Modal	1.5 GPU-minutes a video: about $570 a month on community RTX 4090s, or $1,600 on RunPod serverless L4s
Bandwidth from Bunny	Standard Asian rate: about $2,000 a month	Volume network: about $330 a month, or about $150 with a 720p copy
Copying production videos into R2	About $1,000 more every month, because each month adds 67 TB that keeps costing	Nothing. Workers read from Bunny and keep only results
Reels stored as MP4 files	About 10 TB more every month, about $150 more each month, plus encoding	Clip lists played over PickleCam's video. MP4s only on request
Human review of every game	750 hours a day, impossible	Review only in the beta, then spot checks of about 20 games a day
Storage of results, job queue, monitoring	Small	About $20 to $50 a month
Total	**About $13,000 a month in month one, and growing**	**About $900 to $2,000 a month, roughly $0.007 to $0.015 a video**

The optimized estimate is a target, not a measurement. Costs grow in proportion to volume: at 9,000 videos a day the optimized pipeline costs about $1,800 to $4,000 a month, and at 22,500 a day about $4,500 to $10,000. That makes cost per video the number to keep down.

11.5 What was too expensive, and how the plan fixes it

These cuts are M4 work. The prototype processes every frame and adds a shortcut only if runs get too slow for us to iterate (ADR 0018).

GPU compute was the biggest line, at about $0.07 a video. The plan cuts it in five ways, and M2 and M3 report GPU-seconds per video by stage in the eval output as information.

1. A cheap first pass at 1 frame per second finds the play segments, the player count and very short or singles recordings. Full processing runs only on game segments of doubles recordings of 5 minutes or more.
2. The ball tracker runs only during rallies, which are roughly 40% of a game. Player detection and pose run at about 5 frames per second, with tracking filling the frames between.
3. Video decoding moves to the GPU's video decoder (NVDEC) on the Linux workers.
4. Models run in half precision with TensorRT, in large batches, and stay small: RF-DETR's smaller sizes and a lightweight ball network.
5. M4 benchmarks the same 100 games on community RTX 4090s (RunPod, Vast.ai), RunPod serverless L4s, Modal and AWS spot instances, and a fixed-price dedicated GPU server such as Hetzner's at about €214 a month. The cheapest option that finishes each day's videos reliably wins. Batch jobs can retry an interrupted video, so cheap interruptible GPUs are fine.

Bunny bandwidth was the second biggest line, at up to $2,000 a month. It lands on PickleCam's Bunny bill. Appendix B asks PickleCam who pays it and for a separate Bunny pull zone on the volume network for DinkData's downloads, which costs $0.005 per GB instead of $0.03, with our workers placed near one of its locations. No library has a 720p version today. A 720p copy would more than halve the bytes, so we ask for one only if M2 shows that 720p keeps the ball tracker accurate enough, since the far ball is already small.

Copying every production video into R2 would add about $1,000 every month, forever. In production, workers read each video from Bunny while PickleCam still has it, process it, and keep only the results. `events.json`, `stats.json`, `highlights.json` and the report take about 0.5 MB a video. Per-frame tracks are kept for 30 days for debugging and then deleted. R2 keeps full videos only for the gold sets and training data.

Storing every reel as an MP4 would add about 10 TB every month. A reel is a clip list in `highlights.json`. A web player plays the clips from PickleCam's own video and draws the player marker from the tracking data, so a reel costs no encoding and no storage. When a player taps download or share, a short serverless job renders the MP4 and keeps it for 30 days. Reels work as long as PickleCam keeps the video, which is 14 days, plus any MP4s players render.

Reviewing every game by hand can't scale. The beta keeps full review. In production, confidence decides what ships, owners spot-check a sample of about 20 games a day, and players can report a wrong stat.

M4 ends with a measured cost per video. The target is under $0.01 a video for compute plus bandwidth. If the measurement misses it, the next steps are distilling the ball tracker into a smaller model, lowering the player-detection rate further, and pricing a fixed fleet of dedicated GPU servers.

12. Repository and development workflow

12.1 Layout and tooling

```
src/dinkdata/  contracts data video court ball players events serves stats report highlights
eval studio, cli.py
studio-ui/     contracts/ (generated)    tests/ (synthetic fixtures)
docs/          DESIGN.md and its PDF, STATS COURT DATA LABELLING EVAL OPERATIONS KNOWN-ISSUES,
adr/, agents/, diagrams/
GLOSSARY.md  CLAUDE.md  README.md  LICENSE (AGPL-3.0)  pyproject.toml  uv.lock  Makefile
.github/
```

`DESIGN.md` owns the product, the architecture and the roadmap, and the PDF is generated from it. Topic docs hold the detail. `STATS.md` is the authority for stat code. `GLOSSARY.md` and `docs/adr/` follow the conventions in Marti's PR #26. If a PR makes a topic doc disagree with `DESIGN.md`, it updates both.

CODEOWNERS follows the split in section 5. Robby owns `data`, `video`, `court`, `ball`, `events`, `stats` and `report`. Marti owns `players`, `serves` and `highlights`. Both own `contracts`, `eval`, `studio`, `docs`, `.github` and `CLAUDE.md`.

The core dependencies are numpy, pydantic, pyarrow, polars, typer, httpx and boto3, enough for stats, contracts and eval. The `cv` extra adds torch, torchvision, rfdetr, rtmlib, supervision, av, opencv-headless, scipy and scikit-learn. The `cpu` and `cu126` extras pick the PyTorch build, and the `studio` extra adds fastapi, uvicorn and jinja2. This list is the target. Each library joins `pyproject.toml` with the ticket that first uses it, and new dependencies need the other owner's approval.

Pre-commit runs ruff, gitleaks, a large-file block, notebook stripping, a block on commits to `main` and the `frame / 30` check. CI runs lint, pyright in standard mode, tests and the eval smoke test on Ubuntu for every PR, and on macOS weekly. Tests marked `gpu` or `data` run on a cloud GPU.

12.2 Building with Matt Pocock's skills

Marti's PR #26 configures the repo for mattpocock/skills v1.3.1, a set of Claude Code skills for building in small reviewed steps. Each owner installs them with `claude plugin install mattpocock-skills@claude-plugins-official`. A feature goes through these steps.

1. `/grill-with-docs` interviews you about each decision before any code. It records settled terms in `GLOSSARY.md` and decisions in `docs/adr/`.
2. `/to-spec` writes the spec as a GitHub issue.
3. `/to-tickets` splits the spec into small tickets, linked by GitHub's issue dependencies.
4. `/implement` takes one ticket per fresh session and works test-first.
5. `/code-review` runs two reviewers, one checking the spec and one checking coding standards.
6. `/pr` opens the pull request with our template. The other owner approves and squash-merges.
7. `/retro` records what to change next time.

Large efforts can use `/wayfinder`, which keeps one map issue with child tickets. Issues carry one triage label (`needs-triage`, `needs-info`, `ready-for-agent`, `ready-for-human` or `wontfix`), plus `owner:`, `area:` and `phase:` labels, where the phase is the milestone, M0 to M4. Labelling tasks are `ready-for-human`, and coding tickets are `ready-for-agent` once specified.

This document is the top-level spec. Each milestone section becomes a `/to-spec` issue, `/to-tickets` breaks it down, and separate Claude Code sessions run `/implement` on the tickets. `CLAUDE.md` keeps the rules every session follows: the layout, pydantic contracts, timestamps, floor-only mapping, no loading at import time, `pathlib`, the privacy rules, eval tables, and asking before adding dependencies or changing contracts. It stays under 150 lines and keeps the agent-skills section from PR #26. Claude Code settings block reading `.env` and pushing to `main`. All docs follow the unslop writing rules, which ban AI filler words, em dashes, inline-header lists and vague claims.

13. Licensing

The repo uses AGPL-3.0. Every model and library we depend on has a permissive license, so a closed PickleCam version would need no paid license. PickleCam owns the code, and only the two owners contribute, so it can relicense its own work. Outside contributors would first sign a CLA.

Component	License	Note
RF-DETR, sizes Nano to Large	Apache-2.0	The XL and 2XL sizes use a different license (PML), so we don't use them
RTMPose, through rtmlib or MMPose	Apache-2.0	
ByteTrack, through supervision	MIT	
OSNet, through torchreid	MIT	Appearance model for re-identification
PyTorch, PyAV, NumPy, SciPy, scikit-learn, uvicorn, Jinja2, httpx	BSD	PyAV bundles an LGPL build of FFmpeg, which is fine for a library
OpenCV, pyarrow, boto3	Apache-2.0	
pydantic, polars, typer, FastAPI, rclone	MIT	
CVAT, used as a separate tool	MIT	Not part of our code

We switched from Ultralytics YOLO before training anything. Ultralytics is AGPL-3.0 and says the license also covers models trained with it, so a closed version would have needed its paid Enterprise license. While we only use pretrained models the switch costs almost nothing, and switching after training would mean redoing work. We reuse no code from `legacy/` or TrackNetV3, and we write the ball tracker from the papers.

14. Roadmap, setup phase and first issues

14.1 Milestones

There is no deadline. Each milestone ends with a measurable result. M0 to M3 build the prototype we show PickleCam's CEO, and M4 starts once PickleCam approves it (ADR 0018). Work that only serves integration or production cost waits for M4.

Milestone	Delivers	Done when
Setup	Section 14.2	The docs and package layout are merged and the issues are ready
M0 Foundations	Contracts v1, the endpoint spec sent, 150 hand-picked and hand-downloaded games prepared with `dink ingest` and stored in R2, cloud accounts with guardrails, stats v1 on hand-written logs and CVAT ball clicks with a converter (Robby). The Docker image, the highlight renderer on a hand-written rally list, the minimal rally labeller and the player spike with partner-identity accuracy (Marti). Rally-level labelling starts	Contract tests pass, 150 games with timestamps are in R2, stats tests cover every v1 rule, a reel renders from a hand-written list, and partner-identity accuracy is measured for near and far teams
M1 Ground truth	The calibration tool and calibrations, the camera-moved check, the studio with its events mode, the rest of the rally-level gold set and the first full-event games, the eval command with baselines and coverage per setup, and a first end-to-end version that produces a report and a reel	`dink run` produces a report and reels, `dink eval` prints every metric, and agreement is measured
M2 Serve states and first reels	Serve states, game mode, rallies, the serve sequence with rally winners and game boundaries, and rally-based reels, designed around the measured partner-identity accuracy (Marti). Ball tracker v1 (Robby)	The serve-state, rally-winner and rally-reel gates pass, and the ball reaches 90% at 10 px
M3 Ball stats and full reels	Silver labels, the spotter, the rally grammar and attribution, bounces and in/out, stats 2 to 5 and the landing map (Robby). Reels with winners and aces (Marti)	The stats and events gates pass, with coverage reported
M4 Integration and beta, after PickleCam approves the prototype	Automatic intake (endpoint, `dink catalog sync`, `dink sample`, `dink fetch` and the daily workflow), `dink run` on loose files with automatic camera matching, reels played over PickleCam's video, the play-segment and rally-only speed-ups, the batch runner, review mode, automatic calibration, thresholds, delivery to beta players, retraining, the optimized Linux Docker image, and the provider benchmark on 100 games	A week of reviewed games with fewer than 1 stat in 10 changed, and cost under $0.01 a video

Later work includes dink success rate, an all-shots heat map, unforced errors, forehand versus backhand, singles, the WASB comparison, self-training, production in the cloud, a web app and the PickleCam integration.

14.2 The setup phase, done by this session after approval

Setup prepares the repo and GitHub so that separate Claude Code sessions can build the milestones from issues. Setup builds no pipeline features. It works on branches, Robby confirms every push and GitHub change, and Marti reviews the PRs. Setup starts after Marti's PR #26 is merged, and builds on it.

PR A covers the design and the docs.

- `docs/DESIGN.md`, its PDF and the diagrams.
- `GLOSSARY.md`, seeded from section 4, and ADRs in `docs/adr/` for the decisions in Appendix A.
- Short topic docs: `STATS.md`, `COURT.md`, `DATA.md` with the endpoint spec and privacy rules, `LABELLING.md`, `EVAL.md`, `OPERATIONS.md` and `KNOWN-ISSUES.md`.
- A new `CLAUDE.md` under 150 lines that keeps the agent-skills section, and a new `README.md`.
- `docs/agents/issue-tracker.md` updated for the new ownership split and the M0 to M4 phase labels.
- `docs/DinkDataRebuildPlan.pdf` and `docs/FAILURE-LOG.md` removed, with the prototype's lessons kept in ADR 0001.

PR B sets up the package layout.

- The `legacy-prototype` tag, then `legacy/` deleted.
- Empty `src/dinkdata/` packages with docstrings, including `highlights`.
- A `dink` CLI, renamed from `dink-cv`, whose subcommands (`catalog`, `sample`, `fetch`, `calibrate`, `run`, `eval`, `studio`, `report`, `highlights`) print "not built yet".
- Split dependencies, the AGPL-3.0 license, a clean `.gitignore` and the Makefile.
- CI on Ubuntu for every PR and macOS weekly, pyright standard and the `frame / 30` check.
- Updated CODEOWNERS, PR template and Claude Code settings.
- The existing tests, adapted. The old `events.schema.json` stays until contracts v1 replaces it.

On GitHub, setup closes the 16 old issues with a note pointing here. It creates the labels (triage, owner, area including `area:highlights`, phase M0 to M4, `accuracy-change`) and the milestones. It writes the M0 and M1 work as spec issues in the format `/to-spec` produces, splits them into tickets in the format `/to-tickets` produces, with dependencies between them, and adds them to the project board.

Before merging, `make check` passes on Ubuntu and macOS, `dink --help` lists the commands, the PDF renders from the Markdown and no personal data appears anywhere.

14.3 First issues, for the building sessions

#	Issue	Owner	Phase
1	Contracts v1: pydantic models, generated schemas, examples, round-trip tests. Marti writes the players, rallies, score and highlights models, Robby the rest	Both	M0
2	Send the endpoint spec and Appendix B to the backend developer	Robby	M0
3	R2 bucket, keys, rclone and the nightly backup of labels and calibrations	Robby	M0
4	Pick the held-out facilities and download the first games by hand from training facilities	Both	M0
5	`dink catalog sync`, against a recorded response first	Robby	M4
6	`dink sample` with facility hold-out and setup coverage	Robby	M4
7	`dink ingest`: drop audio, timestamps, HMAC id, catalog row and upload for a hand-downloaded game. `dink fetch` and the daily workflow follow in M4	Robby	M0
8	`dinkdata.video`: a frame-accurate reader with timestamps	Robby	M0
9	Stats v1 as pure functions, tested on hand-written logs	Robby	M0
10	CVAT for ball clicks, and a converter to `ball_labels` with a frame-alignment test	Robby	M0
11	A RunPod account and team on Robby's card, with prepaid credit, auto-pay off, a $5 low-balance email and a 4-hour pod timer	Robby	M0
12	The Docker image, tested on a pod, and `OPERATIONS.md` commands	Marti	M0
13	Highlight renderer: cut, mark and join clips from a hand-written rally list	Marti	M0
14	Minimal rally labeller: a local web page that writes `rallies.json`	Marti	M0
15	Spike: RF-DETR, RTMPose and ByteTrack on 5 games, court filter, cut-off feet, and partner-identity accuracy at the serve for near and far teams	Marti	M0
16	Start labelling the rally-level gold set	Both	M0
17	Calibration click tool, solver, shared lens, visibility mask, floor functions	Robby	M1
18	Calibrate the sampled cameras, set the threshold, camera-moved check	Robby	M1
19	Studio server and events mode	Marti	M1
20	Finish the rally-level gold set, label the full-event gold set (4, then 12), the agreement game and foot clicks	Both	M1
21	Eval v1: all metrics, baselines, coverage per setup, intervals, CI smoke test	Both	M1
22	First end-to-end version: ball baseline, rule events, stats and report (Robby), and rallies from serves with a reel (Marti)	Both	M1

The M2 to M4 issues are written when those milestones start.

15. Risks and open questions

Risk	Response
Labelling takes longer than planned, which is likely	Labelling starts in M0 without waiting for the studio: a minimal rally labeller, CVAT for ball clicks and the calibration click tool. 4 full games first, model suggestions, and gold sets that grow over time
The far ball is too small on low cameras	The crop, larger inputs, more labels from low cameras, and leaving stats out with coverage reported

Risk	Response
Partners can't be told apart, or swap ids, and per-player serve stats break	Measure partner identity in the M0 spike for near and far teams before designing M2. Appearance check at each serve, the parity hint only as a tiebreak, and shipping per-player serve stats only for setups that pass
Players break the serving rules, a few times in most games	No output shows the score, rally winners come from local evidence, and rally-winner accuracy is reported separately for games with deviations, as section 6.10 describes
Stacking and other unusual positions	Weak evidence, not rules, measured on the rally-level gold set
The endpoint arrives late	It isn't needed until after the prototype. We download games by hand
Many camera models	Group by facility, share lens parameters, automatic calibration in M4
A camera gets bumped	The camera-moved check from M1
Overfitting to small test sets	Facility hold-out, intervals, results by setup and growing test sets
Agents write plausible but wrong code	Run-time contracts, tests, eval gates and review
Videos expire before we copy them	Download each picked game within 14 days, and after the prototype a daily scheduled fetch
Development cloud bills run over budget	Prepaid credit with auto-pay off, a $5 low-balance email, a 4-hour pod timer, a $50 monthly cap and GPU cost shown in every eval
Cheap GPUs get interrupted mid-run	Checkpoints at least every 30 minutes, and jobs that resume
Personal data leaks into the public repo	Hashed ids, gitleaks, the rules in `CLAUDE.md` and review
Production costs grow with volume, from about 4,500 videos a day	In M4: the optimizations and provider benchmark in section 11.5 and a target under $0.01 a video
Bunny bandwidth costs PickleCam up to $2,000 a month	A volume-network pull zone, as Appendix B asks, and a 720p copy if M2 shows it is accurate enough
A closed PickleCam version is wanted later	Every model and library is permissively licensed, and only the owners contribute, so PickleCam can relicense its own code

These questions are still open.

1. When will the read-only endpoint exist? PickleCam builds it after the prototype.
2. How common are stacking, rule breaks and mid-game end switches? The rally-level gold set will tell.
3. How well do two people agree on hits and bounces? We'll measure it in M1.
4. What reprojection error can we reach? The first 20 calibrations will tell.
5. How do beta players receive reports and reels, through PickleCam with the video or from the owners directly?
6. Is a 70% coverage floor right?
7. Which stats and which kinds of clips do beta players value most?
8. Will PickleCam set up a volume-network pull zone for production? It pays our download bandwidth for the prototype.
9. How fast does PickleCam expect to grow beyond 4,500 videos a day?

16. Sources

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Appendix A: Decision log

These decisions come from the review session of 17 October 2026, Robby's notes and Marti's review.

#	Decision	Reason
1	Standalone first, PickleCam integration later	Prove accuracy before touching production
2	Two v1 priorities: stats (Robby) and per-player highlight reels (Marti)	The owners' notes
3	Ownership follows the priorities (section 5)	Each owner controls the stages behind their priority
4	Every player gets the whole-game report and every reel, and finds themselves by photo	The simplest identity solution
5	Doubles with side-out scoring, games to 11 by default and longer games supported. Superseded by 40	Most games. Singles and rally scoring later
6	Five v1 stats and the landing map. Dinks, unforced errors and forehand versus backhand later	Most wanted and cheapest to build
7	Targets on per-stat counts, events from human agreement, and coverage	A ±5-point target per player couldn't be measured
8	Two gold sets, 12 full games and about 50 rally-level games	Enough games to measure rally winners, serve states and clip edges
9	Report the score as played, and flag the players' rule breaks. Superseded by 40	Recreational players make serving mistakes
10	Human review of every beta game	Trust while the models improve
11	Minimal spending, no contractors, we label ourselves	Budget
12	Cloud GPUs from the start in one Docker image, the cheapest GPU that gives the same result, a $50 monthly development budget, and production chosen by measured cost in M4	One environment, cloud skills learned early, no switch before launch
13	Apache and MIT models for players (RF-DETR, RTMPose, ByteTrack), replacing Ultralytics YOLO, and the repo stays AGPL-3.0	Switching costs almost nothing while we only use pretrained models, and a closed PickleCam version needs no paid license. Marti's review
14	R2 storage with a local copy, and a read-only endpoint instead of Bunny keys	Independent of production, and Bunny keys can delete videos
15	No audio	Too noisy at multi-court venues
16	Start from scratch and remove `legacy/`	Its labels and checkpoints are lost and its code doesn't fit
17	Our own studio for labelling and review	Pickleball-specific, writes our contract, becomes the review tool
18	Matt Pocock's skills for all development	Marti's PR #26
19	CI on Ubuntu for every PR, macOS weekly	Matches our Macs and the Linux GPUs
20	This session does setup only, and separate sessions build	Robby's request
21	Partner identity measured in the M0 spike before M2 is designed	Marti's review: the riskiest assumption
22	Stats must beat a median-count baseline, and coverage is reported per camera setup	Marti's review
23	Labelling starts in M0 with a minimal rally tool and CVAT, before the studio	Marti's review: labelling is the critical path
24	Production target under $0.01 a video, measured in M4	4,500 videos a day and growing

#	Decision	Reason
25	Production reads videos from Bunny and stores only results	Copying videos would add about $1,000 a month every month
26	Reels are clip lists played over PickleCam's video, with MP4s on request	Storing every reel would add about 10 TB a month
27	Full human review only in the beta, then spot checks	Reviewing 4,500 games a day would take 750 hours
28	Labels made in a browser store time as `t`, and `frame` is filled later	A browser player reports `currentTime` but can't land on exact frames. Contracts v1 grilling
29	Hand-downloaded games get their `video_id` from the same HMAC function as the catalog	Labels made before `dink catalog sync` exists keep their ids. Contracts v1 grilling
30	One read-only call, a recording list with each MP4's download link, replaces the list and links calls	The dashboard already has the fields, and PickleCam owns the link format. ADR 0017
31	Fetch prefers Bunny's 1080p MP4 over the original upload	Only 1 of 6 libraries keeps originals, they are about 4 times larger, and the MP4 keeps dropped-frame gaps. ADR 0017
32	An accurate prototype for PickleCam's CEO comes first. Automatic intake, delivery and cost work come after it	PickleCam's backend developer, 20 October 2026. ADR 0018
33	DinkData is a standalone app until PickleCam approves the prototype, and M4 becomes Integration and beta	PickleCam pays only for accurate, reliable stats. Prototype grilling, 21 October 2026
34	The prototype renders a reel MP4 for every player. The web player over PickleCam's video waits for M4	The web player depends on PickleCam's systems. ADR 0012
35	The prototype processes every frame. Speed-ups and the cost target wait for M4, and eval reports GPU time as information	Shortcuts add code and can cost accuracy. ADR 0018
36	`dink run` takes an ingested game and finds its camera in the catalog. Each new camera gets a one-time hand calibration	Loose files and automatic camera matching and calibration are M4. ADR 0018
37	R2 stays as the development store, and `dink run` reads models from a local folder	Labels and games need one shared, backed-up place. ADR 0006
38	A training run that measurably improves accuracy can exceed the $50 monthly budget if both owners agree first	Accuracy wins over saving money
39	M0 uses RunPod only, with prepaid credit as the cap and a 4-hour timer on every pod. Modal waits for batch eval and Vast.ai for the M4 benchmark	One account until a ticket needs more. RunPod can't stop idle pods
40	No output shows the score or any number derived from it. The pipeline may track the score in memory as weak evidence, gold labels keep it, and v1 covers doubles under any scoring system	PickleCam's CEO, 21 October 2026: players often mis-score, and a score that differs from theirs looks wrong. ADR 0019

Appendix B: Questions for the PickleCam backend developer

We sent these on 20 October 2026, each with a short reason, and the backend developer answered the same day. We answered the other questions from an earlier draft ourselves from the facility dashboard. `docs/DATA.md` records both.

1. Can you add one read-only call, protected by a token, that lists recordings across all facilities with the fields in section 8.2 and no personal data? Not before the prototype.
2. Which camera models are installed, and at which facilities? Hard to find out, and training on several lenses helps. We calibrate each camera from its video anyway.
3. Can you turn on MP4 fallback for the one library we found with it off? Our report was wrong. The MP4 hadn't finished encoding when we checked.
4. Who pays the Bunny bandwidth for our downloads, and can you add a pull zone on the volume network that only we use? PickleCam pays, and cost work waits until after the prototype.